Acyclovir-Induced Neurotoxicity: Concentration-Side Effect Relationship in Acyclovir Overdose

WALTER E. HAEFELI, M.D., RONALD A.Z. SCHOENENBERGER, M.D., PHILIPP WEISS, M.D., RUDOLF F. RITZ, M.D., *Basel, Switzerland*

PURPOSE: To investigate the concentration–side effect relationship in a patient with severe acyclovir-induced neurotoxicity and to summarize the information available in the literature about central nervous system side effects due to acyclovir.

METHODS: Repeated blood samples were drawn in a patient with severe acyclovir overdose who developed coma and nonoliguric renal failure. The acyclovir levels measured by radioimmunoassay were related to the level of consciousness.

RESULTS: We measured the highest acyclovir serum levels reported so far (229.9 μmol/L = 51.8 mg/L). Impairment of consciousness developed with a remarkable temporal delay of 24 to 48 hours after occurrence of peak serum concentrations and resolved with a comparable delay after reaching the therapeutic range (anticlockwise hysteresis). Six days after discontinuation of the drug, central nervous system symptoms had resolved, and, 4 days later, renal function returned to pretreatment values.

CONCLUSIONS: The observation that neurotoxicity developed with a delay of 24 to 48 hours after acyclovir peak serum concentrations could explain the wide range of acyclovir levels reported in similar cases. Single drug level measurements may therefore be of little diagnostic value. Since toxicity develops with a remarkable delay, early removal of the drug (by hemodialysis) could possibly prevent central nervous toxicity.

From the Department of Internal Medicine, Division of Medical Intensive Care, University Hospital (Kantonsspital), Basel, Switzerland.

Dr. Haefeli's research is supported by the Swiss National Science Foundation.

Requests for reprints should be addressed to Walter E. Haefeli, M.D., Stanford University Medical Center, Division of Clinical Pharmacology, Room S-169, Stanford, California 94305-5113.

Manuscript submitted December 4, 1990, and accepted in revised form May 9, 1991.

Acyclovir is a usually well-tolerated antiherpetic agent. Vomiting and hypotension after intravenous administration and, occasionally, nonoliguric renal failure in dehydrated patients are well-known untoward effects [1]. Confusion, hallucinations, seizures, and coma are much rarer and have been reported in particular in immunocompromised patients [2]. Since there is a wide overlap of serum concentrations in patients with and without neurologic side effects, the relation between central nervous system effects and acyclovir serum concentrations remains unclear [2–7]. Until now, there have been no data available describing the time course of acyclovir serum concentrations in conjunction with the neurologic disturbances. Although there is some evidence that occurrence of acyclovir toxicity is related to its serum levels, no clear concentration–side effect relationship has been established so far. We describe a patient with exceptionally high acyclovir serum levels in whom close monitoring of acyclovir serum concentrations and of her neurologic state allowed investigation of the concentration–side effect relationship. Twenty-three other cases of acyclovir-induced neurotoxicity described in the literature are discussed, and their results are compared with the findings in this case.

CASE REPORT

A previously healthy 77-year-old woman weighing 59.5 kg was treated with intravenous acyclovir (1 g three times daily in normal saline over 2 hours) for severe ophthalmic herpes zoster. At admission, signs of central nervous system involvement were absent. Renal function was mildly impaired (urea 12.7 mmol/L [normal: 3 to 7.8 mmol/L]; creatinine 96 μmol/L [normal: 62 to 124 μmol/L]). Serum electrolytes, glucose, and blood cell counts were normal. Two days later, her neurologic condition rapidly deteriorated. Impaired consciousness, dysarthria, myoclonus, hyperactive tendon reflexes, and a left-sided Babinski's sign developed (**Figure 1**). A computerized tomographic scan of the brain was normal; electroencephalography showed moderate generalized disturbances without focal abnormalities or signs of encephalitis. Cerebrospinal fluid was clear with 30 cells/μL, a glucose level of 4.3 mmol/L,

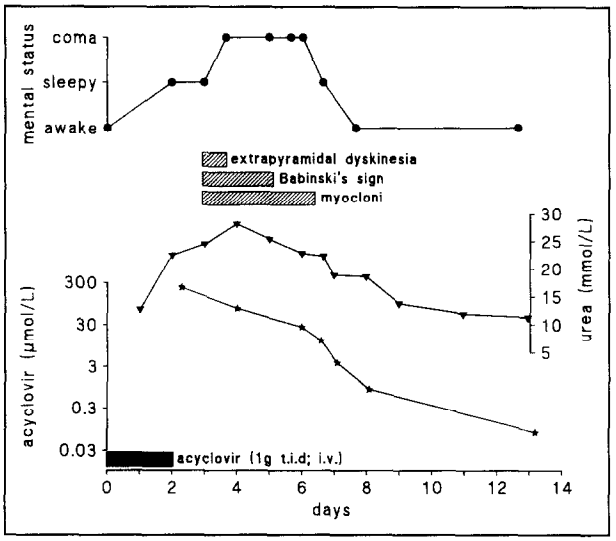


Figure 1. Time course of state of consciousness (•), renal function (▲), and acyclovir serum concentrations (*) in a patient with acyclovir-induced neurotoxicity.

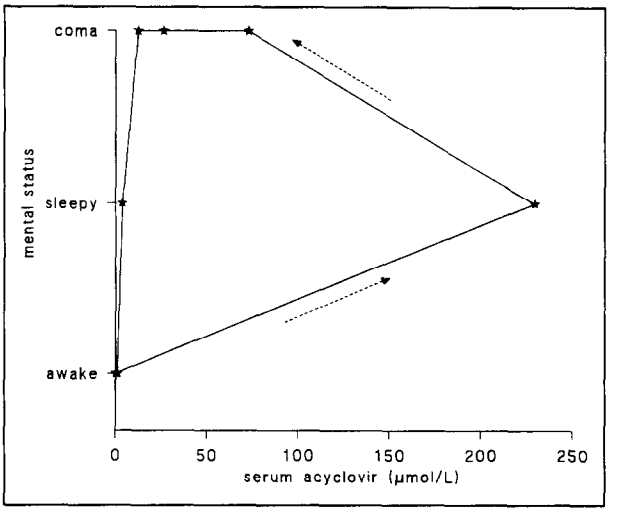


Figure 2. Anticlockwise hysteresis of acyclovir toxicity in a patient with acyclovir neurotoxicity (relation of acyclovir serum concentrations [µmol/L] to the mental status of the patient).

and a lactate level of 2.7 mmol/L (normal: 0.18 to 0.41 mmol/L). Cultures for bacteria and fungi were negative. Her serum creatinine and urea levels had risen to 394 µmol/L and 22.5 mmol/L, respectively. Assuming acyclovir neurotoxicity and nephrotoxicity, treatment was discontinued after a total dose of 6 g. No psychotropic co-medication had been administered, and amoxicillin (1 g three times daily), which had been started because of suspected bacterial superinfection of the eye, was discontinued after the second dose (Day 3). Nevertheless, the patient's mental status further deteriorated over the next 2 days, and she became unresponsive on Day 4. At that time, omeprazole (20 mg/d; Days 1 to 4), which had been administered because of a history of hiatus hernia with esophagitis, was discontinued, and moderate doses of furosemide (40 mg/d) remained the only other medication administered (Days 3 to 5). The serum creatinine and urea levels rose to a maximum of 514 µmol/L and 28.2 mmol/L, respectively (creatinine clearance 8.5 mL/min, urinary output 2,900 mL/d, no urinary crystals found), then gradually normalized within 10 days. During the following days, neurologic symptoms gradually disappeared (Figure 1). Six days after discontinuation of acyclovir therapy, neurologic symptoms had completely resolved. A second computerized tomographic scan was still normal, and electroencephalographic findings had improved.

A summary of acyclovir serum concentrations collected during the investigation period and kindly determined by Wellcome Foundation Ltd., Beckenham, United Kingdom, is depicted in Figure 1. A double antibody radioimmunoassay with a monoclonal antibody to acyclovir (detection limit less than 0.03 µmol/L) was used. Extremely high acyclovir levels with a peak value of 229.9 µmol/L (51.8 mg/L) were found. A marked delay between the serum concentration-time profile and onset and resolution of the neurologic disturbances was observed (anticlockwise hysteresis; **Figure 2**).

TABLE I

Frequency of Central Nervous System Effects in 24 Patients With Acyclovir Toxicity*

Effects	%
Tremor, myoclonus	58
Confusion	50
Agitation	38
Lethargy	25
Hallucinations	25
Extrapyramidal symptoms	21
Clouding of consciousness	17
Dysarthria	17
Unilateral, focal symptoms	13

*[2–16] and present case.

COMMENTS

Acyclovir is mainly eliminated by the kidney, and dose adjustment in patients with impaired renal function is recommended. In the past, several cases of acyclovir-induced neurologic side effects have been published [2–16], and, in several larger study populations, neurologic side effects have been observed [17–19]. Symptoms of neurotoxicity (**Table I**) usually occurred within the first 24 to 72 hours of

treatment. In more than two thirds of the cases, marked deterioration of renal function or pre-existing end-stage renal disease was noted, but toxicity has also been reported after up to 120 days of treatment without changes in renal function [15]. Resolution of renal failure and neurotoxicity is usually complete and occurs within 2 to 7 days.

With the exception of renal failure, no definite predisposing factors for neurotoxicity have been defined. Since the prevalence of severe herpes infections is higher in immunocompromised patients, it is not surprising that most of the patients reported until now were transplant recipients [2,9,17–19] or patients with tumors [3–5,7,11,13,15,18]. Neurotoxicity has also been reported in immunocompetent patients without underlying disease or renal insufficiency [10,12]. Several facts argue for a relation of acyclovir serum concentrations to neurologic side effects, but so far no clear relationship has been established.

(1) Although neurotoxicity has also been reported after oral dosing [8,15], most side effects were recognized after intravenous administration where higher doses are usually used.

(2) In most cases, toxicity occurred either at the beginning of therapy before or shortly after achieving steady-state conditions or after increasing the dose regimen [2].

(3) Improvement of neurologic symptoms after discontinuation of the drug or after its removal by hemodialysis [4,5,8] further argues for a concentration-dependent phenomenon.

(4) Other acyclovir side effects are correlated with its serum concentration. In all patients with peak concentrations above 111 μmol/L (= 25 mg/L), Bean and Aeppli [20] found moderate to severe side effects, and all patients with moderate or severe side effects and lower peak serum concentrations had impaired renal function. This finding supports the suggestion that drug toxicity probably correlates better with factors affecting drug elimination such as the area under the concentration time curve or the elimination half-life than with single (peak) concentration measurements.

The Current Case

The temporal association of acyclovir administration and neurologic signs resembling clinical symptoms and electroencephalographic findings in previously published cases [2–19], as well as the absence of another major cause, strongly suggest a drug-related side effect in the patient presented. Uremic encephalopathy may also produce many of the neurologic symptoms found in our patient, and acute renal failure may have influenced not only acyclovir pharmacokinetics but also the patient's susceptibility to drug-induced neurotoxicity. It is, however, unlikely that the only moderate elevations of serum urea can fully explain her neurologic presentation since patients with overt uremic symptoms usually have blood urea nitrogen concentrations greater than 36 mmol/L [21].

In this patient, the high acyclovir dose regimen associated with acute nonoliguric renal failure resulted in one of the highest acyclovir serum concentrations published so far (229.9 μmol/L = 51.8 mg/L) with elevated serum concentrations over a period of several days. As in most other reported cases, neurologic side effects occurred with a delay of 1 to 2 days after the maximum acyclovir concentration. Although no other drugs were administered at that time and acyclovir concentrations had returned to therapeutic levels (herpes simplex: 1.4 to 4 μmol/L; herpes zoster: 3.5 to 6.9 μmol/L; [1]), impaired consciousness persisted for another day. This remarkable temporal shift between the serum concentration-time profile and the occurrence and resolution of side effects (anticlockwise hysteresis, Figure 2) may be the main reason for the wide overlap of serum concentrations reported in patients with and without neurologic side effects [2–5,10]. Further deterioration after discontinuation of the drug has already been reported by other investigators [4,12]. Since acyclovir is almost entirely eliminated unchanged by the kidney and there are no active metabolites known, the most probable explanation for hysteresis in this patient is that equilibration between blood and brain is very slow. This is in accordance with the case reported by Bataille *et al* [3] in which after completion of hemodialysis acyclovir concentrations in the cerebrospinal fluid were twice as high as in the serum, indicating that no immediate exchange between blood and the central nervous system had taken place.

Since acyclovir-induced renal failure frequently precedes neurotoxicity and equilibration with the central nervous system seems to be slow, the drug should immediately be discontinued in these cases, and hemodialysis should be discussed. Early removal of the drug might prevent central nervous side effects if high serum concentrations are expected (or measured), and equilibrium with the central nervous system is not yet achieved.

In suspected acyclovir neurotoxicity, single drug level measurements are of limited diagnostic value. Acyclovir neurotoxicity may (still) be absent even if excessive serum concentrations are measured, and it cannot be excluded in the presence of a "therapeutic" drug level. For development of acyclovir neurotoxicity, the serum concentrations of the pre-

ceding days seem to be crucial, not the actual serum levels.

ACKNOWLEDGMENT

We wish to thank B. Truog, M.D., of Wellcome, Switzerland, and Miss C.D. Smith and Miss E. Starns of Wellcome, United Kingdom, for determination of acyclovir serum concentrations.

REFERENCES

1. Dorsky DI, Crumpacker CS. Drugs five years later: acyclovir. Ann Intern Med 1987; 107: 859–74.
2. Wade JC, Meyers JD. Neurologic symptoms associated with parenteral acyclovir treatment after marrow transplantation. Ann Intern Med 1983; 98: 921–5.
3. Bataille P, Devos P, Noel JL, Dautrevaux C, Lokiec F. Psychiatric side effects with acyclovir [letter]. Lancet 1985; 2: 724.
4. Spiegal DM, Lau K. Acute renal failure and coma secondary to acyclovir therapy [letter]. JAMA 1986; 255: 1882–3.
5. Johnson R, Douglas J, Corey L, Krasney H. Adverse effects with acyclovir and meperidine [letter]. Ann Intern Med 1985; 103: 962–3.
6. Gill MJ, Burgess E. Neurotoxicity of acyclovir in end stage renal disease [letter]. J Antimicrob Chemother 1990; 25: 300–1.
7. Feldman S, Rodman J, Gregory B. Excessive serum concentrations of acyclovir and neurotoxicity. J Infect Dis 1988; 157: 385–8.
8. Swan SK, Bennett WM. Oral acyclovir and neurotoxicity [letter]. Ann Intern Med 1989; 111: 188.
9. Burguet A, Menget A, Fromentin C, *et al.* Neurologic adverse effects of acyclovir in an allogenic marrow transplant recipient child. Arch Fr Pediatr 1988; 45: 343–5.
10. Jones PG, Beier-Hanratty SA. Acyclovir: neurologic and renal toxicity [letter]. Ann Intern Med 1986; 104: 892.
11. Auwerx J, Knockaert D, Hofkens P. Acyclovir and neurologic manifestations [letter]. Ann Intern Med 1983; 99: 882–3.
12. Cohen SMZ, Minkove JA, Zebley JW, Mulholland JH. Severe but reversible neurotoxicity from acyclovir [letter]. Ann Intern Med 1984; 100: 920.
13. Vartian CV, Shlaes DM. Intravenous acyclovir and neurologic effects [letter]. Ann Intern Med 1983; 99: 568.
14. Tomson CR, Goodship THJ, Rodger RSC. Psychiatric side-effects of acyclovir in patients with chronic renal failure [letter]. Lancet 1985; 2: 385–6.
15. Krigel RL. Reversible neurotoxicity due to oral acyclovir in a patient with chronic lymphocytic leukemia [letter]. J Infect Dis 1986; 154: 189.
16. Fischer A, Fellay G, Regamey C. Toxicité rénale et neurologique de l'acyclovir. Schweiz Med Wochenschr 1990; 120: 1200–3.
17. Saral R, Burns WH, Laskin OL, Santos GW, Lietman PS. Acyclovir prophylaxis of herpes-simplex-virus infections. A randomized, double-blind, controlled trial in bone-marrow-transplant recipients. N Engl J Med 1981; 305: 63–7.
18. Meyers JD, Wade JC, Mitchell CD. Multicenter collaborative trial of intravenous acyclovir for treatment of mucocutaneous herpes simplex virus infection in the immunocompromised host. Am J Med 1982; 73 Suppl: 229–35.
19. Fletcher CV, Chinnock BJ, Chace B, Balfour HH. Pharmacokinetics and safety of high-dose oral acyclovir for suppression of cytomegalovirus disease after renal transplantation. Clin Pharmacol Ther 1988; 44: 158–63.
20. Bean B, Aeppli D. Adverse effects of high-dose intravenous acyclovir in ambulatory patients with acute herpes zoster. J Infect Dis 1985; 151: 362–5.
21. Luke RG. Uremia and the BUN. N Engl J Med 1981; 305: 1213–5.